Jason (Jingxuan) Liu

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EDUCATION

University of California, Los Angeles (UCLA) Expected Jun 2028
B.S. in Electrical Engineering | GPA: 3.97/4.00
Completed Coursework: Signals and Systems, Probability and Statistics, Foundations of Computer Vision
Planned / In Progress: Circuit Theory I & II Honors, Logic Design of Digital Systems, Analog Electronics I, Principles of Feedback Control, Digital Signal Processing, Introduction to Machine Learning

PROFESSIONAL EXPERIENCE

Undergraduate Researcher | UCLA Integrated Sensors Laboratory Jun 2026 – Aug 2026
Summer Undergraduate Research Program

- Engineered an edge-AI wearable for poster-content summarization, coupling low-res detection and high-res capture cameras to stream targeted crops via ESP32-S3 BLE for companion app LLM summarization
- Trained and deployed a task-specific 1.7M-parameter INT8 YOLOv5n detector on Himax WiseEye 2 for poster boxing using PyTorch, achieving 0.833 mAP_{50} across 4 Open Images V7 poster and display classes
- Calibrated cross-camera perspective via OpenCV homography, mapping detection bounding boxes to an OV5640 high-res sensor and tuning exposure, gain, and dynamic cropping to minimize BLE payload size
- Ported an ArduCAM OV2640 8 MHz SPI burst driver to Zephyr with nRF Connect SDK, benchmarking transmission at 0.37 J/image on nRF52840 BLE versus 2.41 J/image on ESP32 Wi-Fi using a Power Profiler Kit II

PROJECT EXPERIENCE

Powertrain Project Engineer | Bruin Racing – Supermileage Vehicle Jun 2026 – Present
Design and simulation phase; hardware build pending

- Designed and simulated a 48 V to 12 V isolated buck converter with active clamp forward converter topology in LTspice for an ultra-efficient hydrogen fuel-cell vehicle, achieving 120 W maximum power at 200 kHz with $< 94\,\mathrm{mV_{pp}}$ ripple ($< 0.8\%$) across eight simulated load points spanning 0.25–10 A
- Simulated a PQ32/30 N97 transformer with 6:4 turns and $L_m = 180\,\mu\mathrm{H}$ in LTspice; varied winding leakage and switching dead times (45–420 ns) to characterize hard switching and MOSFET voltage stress
- Designed mixed-signal sensing front ends for converter protection using an STM32G474, specifying Kelvin sense shunts, unity-gain-buffered dividers, and NTC circuits for on-chip ADC sensing and comparator-driven fault shutdown
- Simulated a two-switch forward-converter alternative that used input-rail clamping to limit switch voltage stress, targeting the Shell Eco-marathon 60 V maximum-voltage requirement

Low-Voltage Subteam Member | Bruin Racing – Supermileage Vehicle Sep 2025 – Jun 2026

- Restored CAN across Adafruit Feather RP2040-based boards at the 2026 Shell Eco-marathon in Indianapolis; used custom MCU firmware to test signal passage and repaired reversed pinouts and damaged wiring
- Designed an STM32F446RE rear-control board for torque, pressure, and acceleration sensing, rear-lighting control, and CAN communication with the low-voltage system of the Supermileage Vehicle using KiCad
- Evaluated Hall-effect and hydraulic sensing before integrating a pedal-mounted strain gauge for brake-light triggering

Micromouse | UCLA IEEE Sep 2025 – May 2026

- Designed and assembled a 2-layer STM32F205 PCB integrating dual H-bridge motor control, magnetic encoder, four-direction IR wall sensing, power regulation, and debug interfaces using KiCad
- Configured STM32 timers for quadrature encoder capture and four-channel motor PWM, implementing a 1 kHz closed-loop PD controller for straight-line motion and 90-degree turns
- Implemented time-multiplexed IR sensing with MOSFET-switched emitters, 12-bit ADC acquisition, and DMA averaging across 128 samples per reading to improve wall-detection stability
- Developed a 16×16 flood-fill maze solver in C with wall maps and breadth-first distance updates; the completed Micromouse detected walls and traversed several maze blocks

SKILLS

Prototyping & Test Equipment: PCB design, oscilloscope testing, soldering (hand, reflow and hot plate), simulation
Microcontrollers & Embedded Platforms: STM32, RP2040, ESP32, nRF52840, Arduino, Himax WiseEye 2
Analog & Mixed-Signal: Op-Amps, Unity-Gain Buffers, ADCs, Signal Conditioning
Protocols & Controls: I2C, SPI, BLE, PWM, PID Control, CAN
Software & Tools: STM32CubeIDE, ESP-IDF, nRF Connect SDK, OpenCV, KiCad, LTspice
Programming: C, C++, Python, MATLAB, R